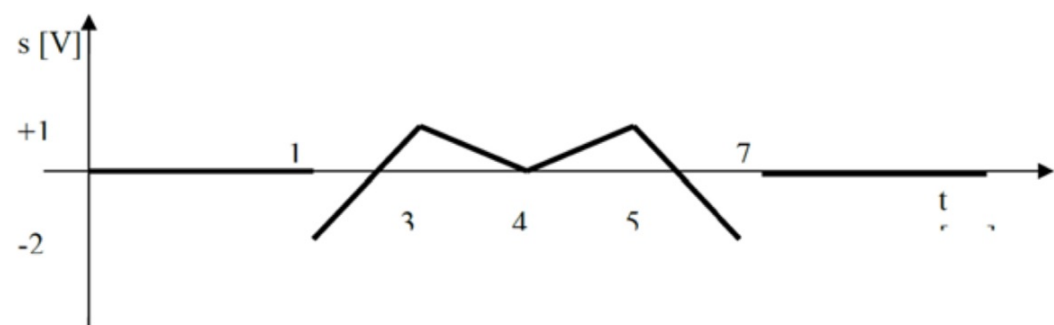


Exercise 3

The signal $s(t)$ represented in the following graph is given:



Write the expression of $s(t)$ and its Fourier transform.

- $S(f)$: ?
- FOURIER SERIES?

Handwritten work showing the decomposition of the signal $s(t)$ into three triangular pulses and the calculation of their Fourier transforms.

Decomposition:

- The signal $s(t)$ is decomposed into three triangular pulses:
 - $S_0(t)$: A green triangle centered at $t=4$ with a height of $5/2$. It is defined by the equation $y = \frac{3}{2}x - \frac{7}{2}$ for $x \in [1, 4]$ and $y = -\frac{3}{2}x + \frac{11}{2}$ for $x \in [4, 7]$.
 - $S_6(t)$: An orange triangle centered at $t=4$ with a height of 1 . It is defined by the equation $y = \frac{3}{2}x - \frac{7}{2}$ for $x \in [3, 4]$ and $y = -\frac{3}{2}x + \frac{11}{2}$ for $x \in [4, 5]$.
 - $S_x(t)$: A yellow inverted triangle centered at $t=4$ with a height of -1 . It is defined by the equation $y = -\frac{3}{2}x + \frac{11}{2}$ for $x \in [3, 4]$ and $y = \frac{3}{2}x - \frac{7}{2}$ for $x \in [4, 5]$.

Fourier Transform Calculations:

For a triangle centered at $t=0$ with base $2a$ and height h , the Fourier transform is $S(f) = h \cdot \text{TRIANG}\left(\frac{f \cdot 2a}{2}\right) + 1$.

Applying this to the pulses:

- $S_0(f) = \left(\frac{3}{2} + 2\right) \text{TRIANG}\left(\frac{f \cdot 6}{2}\right) - 2$
- $S_6(f) = \frac{3}{2} \text{TRIANG}\left(\frac{f \cdot 2}{2}\right) + 1$
- $S_x(f) = -1 \text{TRIANG}\left(\frac{f \cdot 2}{2}\right) + 1$

Final Fourier Transform:

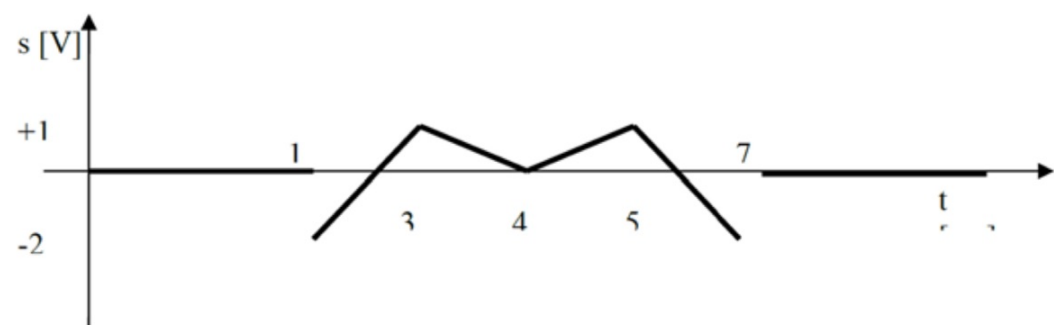
$$S(f) = S_0(f) + S_6(f) + S_x(f) = \left(\frac{3}{2} \text{TRIANG}\left(\frac{f \cdot 6}{2}\right) - 2\right) + \left(\frac{3}{2} \text{TRIANG}\left(\frac{f \cdot 2}{2}\right) + 1\right) + \left(-1 \text{TRIANG}\left(\frac{f \cdot 2}{2}\right) + 1\right)$$

$$= \frac{3}{2} \text{TRIANG}\left(\frac{f \cdot 6}{2}\right) + \frac{1}{5} \text{TRIANG}\left(\frac{f \cdot 2}{2}\right) + 0$$

$$S(f) = \frac{3}{2} \cdot \frac{6}{2} \text{SINC}^2\left(f \cdot \frac{6}{2}\right) e^{-j2\pi f \cdot 4} + \frac{1}{5} \cdot \frac{2}{2} \text{SINC}^2\left(f \cdot \frac{2}{2}\right) e^{-j2\pi f \cdot 4}$$

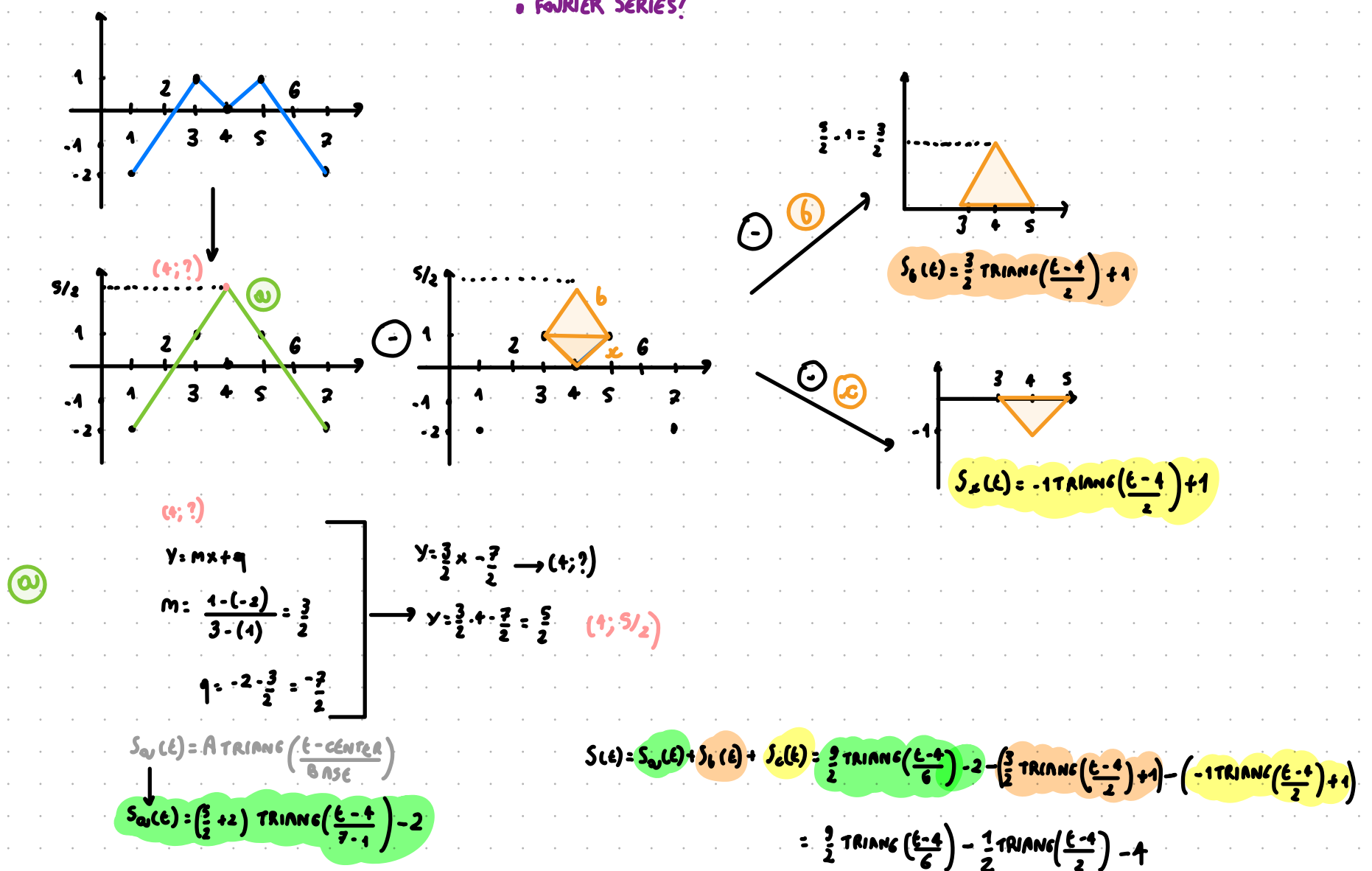
Exercise 3

The signal $s(t)$ represented in the following graph is given:



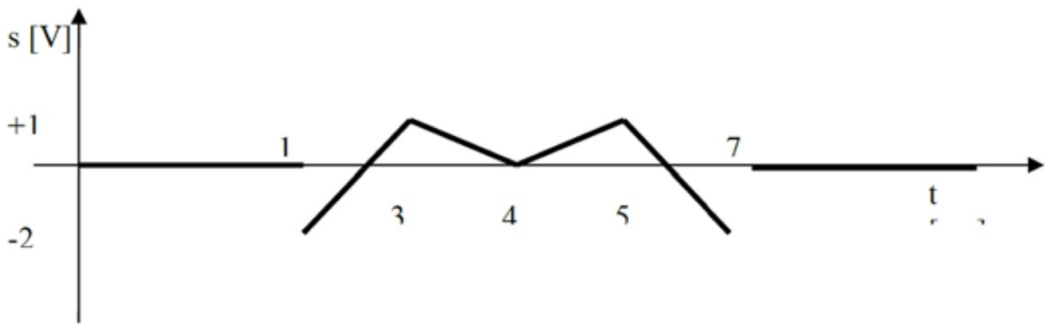
Write the expression of $s(t)$ and its Fourier transform.

- $S(t) = ?$
- FOURIER SERIES?



Exercise 3

The signal $s(t)$ represented in the following graph is given:



WORST

Write the expression of $s(t)$ and its Fourier transform.

ⓐ $s(t) = ?$

$(4; ?)$ $m = \frac{-2 - (+1)}{1 - (+3)} = \frac{3}{2} \rightarrow y = mx + q \rightarrow q = 1 - 9/2 = -7/2 \rightarrow y = \frac{3}{2}x - \frac{7}{2} \rightarrow y = \frac{3}{2} \cdot 4 - \frac{7}{2} = 5/2$

$w(t) = \frac{5}{2} \text{TRIANG}\left(\frac{t-1}{6}\right) - 2$ SHIFT DOWNSIDE

$b(t) = \left(\frac{5}{2} - 2\right) \text{TRIANG}\left(\frac{t-4}{2}\right) + 1$

$x(t) = -1 \text{TRIANG}\left(\frac{t-4}{2}\right) + 1$

$s(t) = w(t) + b(t) + x(t) = \frac{5}{2} \text{TRIANG}\left(\frac{t-1}{6}\right) - \frac{1}{2} \text{TRIANG}\left(\frac{t-4}{2}\right)$

$S(f) = \frac{5}{2} \cdot \frac{6}{2} \text{sinc}^2\left(f \cdot \frac{6}{2}\right) e^{-j2\pi f t} - \frac{1}{2} \cdot \frac{2}{2} \text{sinc}^2\left(f \cdot \frac{2}{2}\right) e^{-j2\pi f t}$